Respiratory Motion-Robust Robotic Ultrasound Acquisitions via Vision-Haptic Fusion Control and 3-D Compensation

Guanglin Cao (曹广林), Danqian Cao (曹丹倩), and Hongbin Liu (刘宏斌)

***Abstract*—Robotic 3-D ultrasound (US) has demonstrated potential for reliable organ volume acquisition. However, respiratory motion during scanning critically reduces volume measurement accuracy, presenting a clinical limitation that affects not only cardiovascular but also all abdominal and spinal US diagnoses. To address this, we develop a robotic system integrating vision-based motion monitoring and interaction control for respiratory motion-robust US acquisition. The motion monitoring module constructs a self-calibrating predictive model that dynamically estimates respiratory motion and predicts the desired contact position. We further propose a vision-haptic fusion motion control framework that integrates predicted position feedback into the force-sensing admittance controller, equipping it with model predictive capabilities. Afterward, to ensure accurate volume measurements, a compensation framework is presented to reduce respiratory motion-induced distortions by reconstructing in a relatively static frame and refining results using anatomical priors. Finally, the whole system was validated using a vessel phantom mounted on a six-axis platform simulating respiratory motion. The experimental results demonstrate that the robotic system ensures interaction stability (0.295-N force tracking accuracy) for US acquisition. The compensation framework significantly improves volume measurement accuracy, achieving a mean spatial distortion of 0.48 mm. These improvements highlight the robotic system's capacity for stable US acquisition and accurate volume measurements across US diagnoses, which are crucial for diagnoses affected by respiratory motion.**

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I. Introduction

ULTRASOUND (US) imaging has become a widely used diagnostic modality for visualizing and quantifying internal tissue diseases due to its noninvasiveness, absence of ionizing radiation, low cost, and real-time capabilities [1]. For instance, abdominal aortic aneurysm (AAA), a localized bulge in the abdominal aorta, necessitates early detection as most patients remain asymptomatic until rupture [2]. Ultrasonography is the primary method used to screen for AAA in primary care because of its high sensitivity (94%–100%) and specificity (98%–100%) [3]. The key objective of this screening is to accurately measure and assess the size of the aneurysm. However, grayscale US images are typically less intuitive than those from computed tomography (CT) or magnetic resonance imaging (MRI). Furthermore, US image quality is highly operator-dependent, relying on visual and tactile expertise, which introduces variability and subjectivity in diagnoses [4]. Recent studies have addressed this limitation by integrating robotic systems to automate probe manipulation, thereby ensuring consistent US acquisition and accurate 3-D volume measurements. The precision of robotic positioning enhances the reliability of quantitative assessments, facilitating the tracking of small volume changes, particularly important in evaluating aneurysm progression under treatment.

Nevertheless, as a continuous contact-based imaging modality, US faces reduced image acquisition stability and volume measurements accuracy due to respiratory motion. This clinical limitation affects not only cardiovascular but also all abdominal and spinal US diagnoses [5], [6]. Specifically, maintaining stable contact between the probe and moving skin is particularly important for US acquisition, but this is a challenging process as patients cannot sustain prolonged breath-holding during scanning. Excessive contact risks patient discomfort or tissue injury, while insufficient contact results in poor-quality US images. Additionally, respiratory motion introduces substantial distortions in volume acquisition, which can significantly affect diagnostic accuracy, especially when diagnosing stenosis by measuring the shape of arteries. During respiration, the probe must follow the up-and-down movement of the skin to maintain contact. Simply stacking 2-D US vessel

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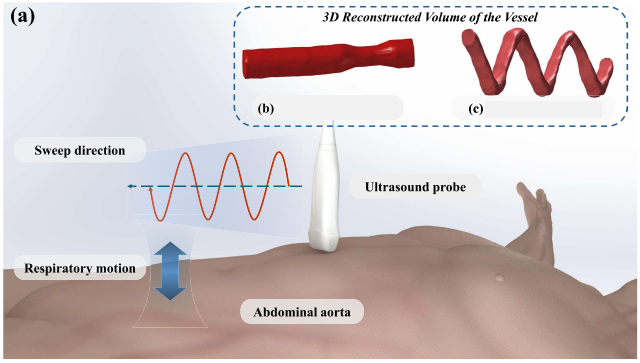


Fig. 1. Illustration of the impact of respiratory motion on US volume measurements. (a) Probe moves up and down along with respiratory motion to maintain stable contact. (b) Reconstructed volume under stationary conditions. (c) Reconstructed volume under respiratory motion.

segmentations in 3-D space based on their poses will cause distortions in the reconstructed volume, as demonstrated in Fig. 1.

This study introduces a robotic 3-D US imaging system designed for stable US acquisition and accurate volume measurements under respiratory motion. To achieve this, the system employs a vision-based motion monitoring module to construct a self-calibrating predictive model that dynamically estimates respiratory motion and predicts the desired contact position. Based on this, we propose a vision-haptic fusion motion control framework that integrates this prediction into a force-sensing admittance controller for stable US acquisition. To correct distortions in the volume measurements, a 3-D compensation framework reconstructs the US volume in a relatively static frame and refines the results using a probability distribution model derived from anatomical priors. Compared to prior studies, the main contributions are listed as follows.

1) A novel respiratory motion-robust robotic 3-D US imaging system. This system facilitates stable US acquisition and precise volume measurements, which is particularly beneficial for clinical applications affected by respiratory motion, such as cardiovascular, abdominal, and spinal imaging.
2) The innovative integration of self-calibrating predictive position feedback into a force-sensing admittance controller. This approach equips the system with model predictive capabilities, enabling it to effectively compensate for respiratory variations.
3) A 3-D compensation framework that combines position compensation and anatomical priors to correct respiratory motion-induced distortions in volume measurements. This represents the first application of anatomical priors for US volume compensation.

To validate its efficacy, the system was thoroughly evaluated on a vessel phantom with a six-axis parallel platform simulating respiratory motion.

II. Related Work

To provide a comprehensive foundation, the related work from the literature is categorized into two main areas: interaction control and US volume compensation. In this section, we review the prior art in these domains and evaluate the applicability of existing methods.

A. Interaction Control

Effective interaction control is fundamental for achieving compliant contact in robotic US systems. To ensure patient comfort and acquire high-quality US images during scanning, various force control methods have been proposed, including PID control [7], stiffness control [8], and Cartesian compliant control [9]. Additionally, impedance control has been implemented to improve the compliance of robotic arms during interactions. For example, Jiang et al. [10] employed an impedance controller to maintain stable contact forces using built-in joint torque sensors in all seven joints of the robotic arm. However, since not all robotic arms are equipped with joint torque sensors, an effective alternative is to install a force sensor at the robot end-effector to detect contact forces and use an admittance controller to maintain stability [11].

Respiratory motion during scanning introduces dynamic changes in tissue stiffness and surface compliance, making fixed-parameter impedance controllers insufficient. To address this, variable impedance control algorithms have been developed. Some strategies employ online optimization to continuously identify the optimal robot stiffness by minimizing the contact force error [12]. Other approaches employ machine learning, such as offline reinforcement learning [13] or confidence-driven adaptive schemes [14], to adjust impedance parameters. However, their high computational demands limit real-time performance, and frequent adjustments to controller parameters can lead to system instability. An alternative strategy to achieve more flexible and adaptive interaction is to incorporate environmental awareness. For instance, in the context of robotic needle insertion, Cho et al. [15] developed a hybrid vision-based variable impedance controller. Their system uses a vision system to estimate respiratory motion velocity and dynamically adjusts the controller's damping parameters to improve compensation. Building on this, Kim et al. [16] directly compared impedance and admittance control for this task, concluding that admittance control offered superior performance in tracking respiratory motion. While these methods effectively compensate for zero-force respiratory motion during discrete, point-based insertions, they are less suitable for continuous US scanning, which demands a stable, compliant contact force over a dynamic surface. Moreover, in contrast to these methods that rely on real-time visual information, our method employs a predictive model to generate forward-looking information, enabling for more effective compensation for dynamic changes.

Robotic systems with environmental awareness can also leverage model predictive control (MPC) for robust interactive control. Vrooijink et al. [17] developed an MPC-based approach to compensate for beating heart motions in robotically actuated delivery sheaths. Their method employed a two-term Fourier series to describe the periodic motion of the aortic annulus, using preoperatively segmented positions from US images to derive the model [17]. However, this reliance on offline data for model fitting makes it unsuitable

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for dynamic motion variations. To address real-time challenges, Duan et al. [18] proposed an admittance-MPC method for robotic lung puncture that integrates image-based disturbance observers to compensate for respiratory motion in real time. However, solving the required quadratic optimization problem in real time increases computational complexity. In contrast, our predictive model can analytically calculate the predicted position, making the computation faster and more efficient.

B. US Volume Compensation

Respiratory motion can substantially degrade the accuracy of volume measurements, especially in cases involving long and complex anatomical structures, such as arteries. Vision-based methods for monitoring the movement of the imaged object have proven to be effective for US volume compensation. Jiang et al. [19] proposed a vision-based robotic US system capable of monitoring the imaged object's motion and automatically updating the sweep trajectory. The visual component employs a depth camera with passive markers to extract manually planned sweep trajectories. By enabling motion detection and compensation, the system can generate high-quality compounded 3-D images. To further address motion-related challenges, Jiang et al. [20] developed another vision-based system that registers the surface point clouds of the object before and after movement using the iterative closest point (ICP) algorithm. This approach facilitates seamless acquisition of a complete 3-D image of the target anatomy. However, these methods interrupt scanning after detecting motion and require trajectory updates before resuming. As a result, they are effective only for discrete rigid motions and cannot accommodate continuous motion. To address this limitation, Heunis et al. [21] introduced a visual-servo-based framework that employs an external motion capture system to continuously adjust the scan trajectory, thereby enabling uninterrupted 3-D reconstruction during periodic motion. However, the reliance on a high-accuracy optical tracking system significantly complicates the clinical workflow and limits its practical feasibility.

Directly extracting respiratory motion patterns from US images to improve the spatial accuracy of volume measurements is another effective method. Fu et al. [22] proposed a system that integrates sparse respiratory signal matching with 2-D US slices for reconstruction. Using manifold learning techniques, respiratory signals are extracted directly from US slices without requiring external devices, enabling efficient alignment of signals across varying positions. Similarly, Velikova et al. [23] introduced an innovative approach that incorporates implicit neural representations (INRs) for breathing-compensated volume reconstruction. Their INR model is trained on tracked US images and corresponding segmentation masks. This process generates a continuous 3-D function that compensates for noisy or missing segmentation frames, thereby producing smoother and more detailed 3-D reconstructions. However, their analysis focused exclusively on assessing volume surface roughness and lacked ground truth comparisons. Our 3-D compensation framework differs by explicitly reconstructing the volume in a relatively static frame and refining the results with anatomical priors. Furthermore, our work includes a thorough evaluation against ground truth to validate its clinical applicability.

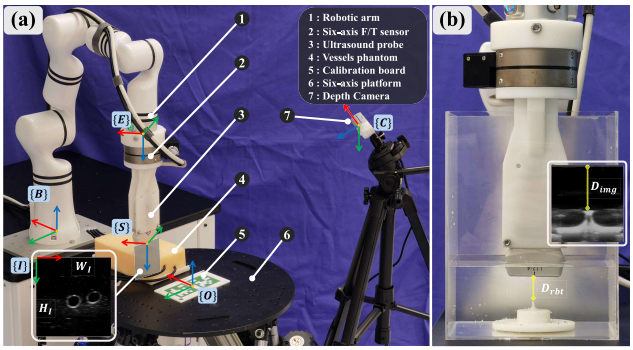


Fig. 2. (a) System setup and coordinate definition. (b) Temporal calibration system setup.

III. Methods

A. Hardware Overview

The robotic 3-D US imaging system consists of two main components: a robot-assisted probe motion system and a vision-based motion monitoring system. The robot system includes an ultralightweight humanoid robotic arm, a medical US device, and a six-axis force/torque (F/T) sensor. In this study, the US device (Pioneer H20, Shenzhen Angell Technology, China), equipped with a linear probe (L12-4), is used to acquire B-mode images. The probe is attached to the F/T sensor (M3815D, Sunrise Instruments, China) via a custom 3-D-printed holder. The F/T sensor is mounted on the end-effector of the robotic arm (RM65-B, RealMan Intelligent Technology, China) to accurately measure the contact force applied to the probe. The vision-based motion monitoring system comprises a depth camera (D455, Intel RealSense Technology, USA) and a custom 3-D-printed AprilTag marker board, which is rigidly connected to the imaged object. The camera is positioned in front of the marker board and provides full six-degree-of-freedom (6-DOF) localization of the imaged object from the captured images. This precise monitoring capability ensures accurate tracking of motion during respiratory movements. The entire system is depicted in Fig. 2(a).

In this study, US images are transmitted to the workstation via a USB stream capture device (CM410, UGreen, China) at a rate of 30 frames/s. Simultaneously, the workstation updates the robotic pose at a frequency of 100 Hz, ensuring temporal consistency between the US images and pose data. The detailed US acquisition parameters are set as follows—imaging depth: 50 mm, focus: 25 mm, frequency: 7.5 MHz, and gain: 66 dB.

B. System Calibrations

1) Spatial Calibration: To ensure accurate US volume measurements, spatial calibration of the system is required based on its mechanical configuration. The coordinate systems used to project pixels from 2-D US images onto voxels in 3-D space

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are as follows: $\{C\}$ denotes the camera frame, $\{B\}$ denotes the robot base frame, $\{E\}$ denotes the robot end-effector frame, $\{S\}$ denotes the probe tip frame, and $\{I\}$ denotes the US image uv frame, as shown in Fig. 2(a). In this configuration, the US image is positioned within the yz plane of $\{S\}$, and the midpoint at the top of the US image is aligned with the origin of $\{S\}$. The homogeneous representation of an image pixel (u, v) can be written as ${}^{I}\mathbf{P} = [0, u, v, 1]^T$. Based on the spatial calibration and the image depth D_I set on the US machine, the relationship between a physical voxel ${}^{B}\mathbf{P}$ and an image pixel ${}^{I}\mathbf{P}$ is mapped as follows:

$$ {}^{B}\mathbf{P} = {}^{B}_{E}\mathbf{T}\,{}^{E}_{S}\mathbf{T}\,{}^{S}_{I}\mathbf{T}\,{}^{I}\mathbf{P} \tag{1} $$

where ${}^{j}_{i}\mathbf{T}$ is the transformation matrix used to transfer the position from frame $\{i\}$ to frame $\{j\}$. The term ${}^{B}_{E}\mathbf{T}$ can be calculated from the robot forward kinematics model, and since $\{S\}$ has the same orientation as $\{E\}$, the combined transformation ${}^{E}_{S}\mathbf{T}\,{}^{S}_{I}\mathbf{T}$ is represented as follows:

$$ {}^{E}_{S}\mathbf{T}\,{}^{S}_{I}\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{D_I}{H_I} & 0 & -\frac{W_I D_I}{2H_I} \\ 0 & 0 & \frac{D_I}{H_I} & L_d \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2} $$

where W_I and H_I represent the width and height of the US image in pixels, respectively, and L_d denotes the physical distance from the tip of probe to the center of the robot end-effector. During scanning, reconstruction is performed by transforming pixels from the US image into 3-D space using (1) and (2).

2) Temporal Calibration: However, data transmission and processing delays can cause a temporal mismatch between the robot kinematics ${}^{B}_{E}\mathbf{T}$ and the pixels ${}^{I}\mathbf{P}$, leading to displacement errors in the reconstructed volume. Temporal calibration involves identifying the relative delay between the two data streams, enabling the synchronized use of matched US images and poses for reconstruction. Based on spatial calibration, physical distances in the real world can be measured with US images. Therefore, by affixing an US-visualized marker in space, the distance from the tip of the probe ${}^{S}\mathbf{P}_0$ to the marker $\mathbf{P}_m$ can be calculated by both the robot kinematics and the US image, denoted as D_{rbt} and D_{img}, respectively, as illustrated in Fig. 2(b)

$$ \begin{aligned} D_{\text{rbt}}(t_1) &= \left\| {}^{B}_{E}\mathbf{T}(t_1)\,{}^{E}_{S}\mathbf{T}\,{}^{S}\mathbf{P}_0 - {}^{B}\mathbf{P}_m \right\|_2 \\ D_{\text{img}}(t_2) &= \left\| {}^{S}\mathbf{P}_0 - {}^{S}_{I}\mathbf{T}\,{}^{I}\mathbf{P}_m(t_2) \right\|_2 . \end{aligned} \tag{3} $$

When the robot performs movements at scanning speed, D_{rbt} and D_{img} are recorded. The average temporal mismatch t_{mis} can be calculated by $t_{\text{mis}} = t_1 - t_2$ when $D_{\text{rbt}}(t_1) = D_{\text{img}}(t_2)$.

3) Hand–Eye Calibration: Hand–eye calibration is essential for transforming the position of the imaged object from the camera frame $\{C\}$ to the robot base frame $\{B\}$. By attaching a calibration board with AprilTag markers to the robot end-effector, the transformation equation is expressed as follows:

$$ {}^{E}_{B}\mathbf{T}\,{}^{B}_{C}\mathbf{T} = {}^{E}_{A}\mathbf{T}\,{}^{A}_{C}\mathbf{T} \tag{4} $$

where $\{A\}$ denotes the calibration board frame, and the transformation ${}^{E}_{A}\mathbf{T}$ can be determined from the mechanical design. By moving the robot to various poses, a set of transformations ${}^{B}_{E}\mathbf{T}_{1,2,\ldots,N}$ and ${}^{C}_{A}\mathbf{T}_{1,2,\ldots,N}$ can be obtained from the robot kinematics model and the AprilTag detection algorithm using the captured images [24], respectively. The transformation ${}^{B}_{C}\mathbf{T}$ can be obtained by optimizing the following equation:

$$ \min_{{}^{B}_{C}\mathbf{T}} \frac{1}{N} \sum_{i=1}^{N} \left\| {}^{E}_{B}\mathbf{T}_i\,{}^{B}_{C}\mathbf{T} - {}^{E}_{A}\mathbf{T}\,{}^{A}_{C}\mathbf{T}_i \right\|^2 \tag{5} $$

where N represents the number of poses. The optimal transformation ${}^{B}_{C}\mathbf{T}$ can be calculated using the least-squares method as follows:

$$ {}^{B}_{C}\mathbf{T} = \begin{bmatrix} {}^{E}_{B}\mathbf{T}_1 \\ {}^{E}_{B}\mathbf{T}_2 \\ \vdots \\ {}^{E}_{B}\mathbf{T}_n \end{bmatrix}^{\dagger}_{4\times 4\ n} \begin{bmatrix} {}^{E}_{A}\mathbf{T}\,{}^{A}_{C}\mathbf{T}_1 \\ {}^{E}_{A}\mathbf{T}\,{}^{A}_{C}\mathbf{T}_2 \\ \vdots \\ {}^{E}_{A}\mathbf{T}\,{}^{A}_{C}\mathbf{T}_n \end{bmatrix}_{4n\times 4} \tag{6} $$

where $\mathbf{A}^{\dagger} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ is the Moore–Penrose pseudoinverse of $\mathbf{A}$.

C. Vision-Based Motion Monitoring

As previously discussed, incorporating motion awareness into a robotic US imaging system can significantly enhance US acquisition stability and volume measurements accuracy. In this study, motion awareness is achieved through a vision-based motion monitoring system capable of determining the 6-DOF localization of the imaged object from captured images. Consistent with established literature [15], [25], this study models respiratory motion as a displacement predominantly in the vertical direction. This approach is justified because vertical movement is the primary factor affecting the stability of US acquisition. With the respiratory motion aligned along the z-axis of the robot base frame, the real-time position of the imaged object, denoted as z_c, is determined by the following equation:

$$ z_c = \mathbf{S}_z\,{}^{B}_{C}\mathbf{T}\,{}^{C}\mathbf{P} \tag{7} $$

where ${}^{C}\mathbf{P}$ represents the homogeneous coordinates of the imaged object's position in the camera frame and $\mathbf{S}_z = [0, 0, 1, 0]$ is a selection matrix. A sinusoidal model is employed to describe the respiratory motion according to the following equation:

$$ z_m(t) = z_0 + A_s \sin\left(\frac{2\pi}{T_s}t + \phi_s\right) \tag{8} $$

where $z_m(t)$ represents the predicted position of the imaged object as a function of time t and z_0 refers to the initial position of the probe tip. The value of z_0 is determined by positioning the probe in contact with the imaged object under stationary conditions using the desired force. Parameters A_s, T_s, and ϕ_s correspond to the amplitude, period, and phase of the motion, respectively. These parameters are initially optimized by minimizing the following cost equation:

$$ f(A_s, T_s, \phi_s) = \sum_{i=1}^{N} \left(z_c(t_i) - \left(z_0 + A_s \sin\left(\frac{2\pi}{T_s}t_i + \phi_s\right)\right)\right)^2 \tag{9} $$

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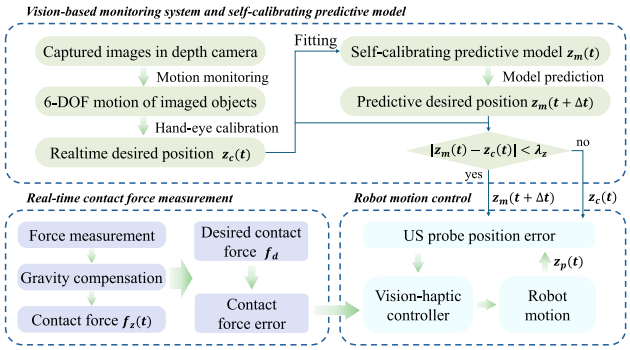


Fig. 3. Our proposed vision-haptic fusion motion control framework, including vision-based motion monitoring system, self-calibrating predictive model, real-time contact force measurement, and robot motion control.

using the Levenberg–Marquardt (LM) algorithm on N sampled data points. During the US acquisition, the model predicts the position of the imaged object. However, variations in respiratory motion frequency may cause discrepancies between predicted ($z_m(t)$) and observed ($z_c(t)$) movements. To address this, a self-calibrating module is constructed using the vision-based motion monitoring system, enabling continuous evaluation of model discrepancies. If the positional error $|z_m(t) - z_c(t)|$ remains within a predefined tolerance threshold λ_z, it indicates that the current movements still align with the model and no parameter updates are required. Conversely, if the error exceeds λ_z, the model acquires $z_c(t)$ and recalibrates the parameters A_s, T_s, and ϕ_s using (9). After each parameter update, the self-calibrating module verifies whether the recalculated positional error lies within λ_z. Once the threshold is satisfied, the model terminates recalibration and resumes predicting desired contact positions.

D. Robot Motion Control

To acquire stable and clear US images, the probe must follow the respiratory motion of the imaged object. However, the conventional force-sensing admittance controller lacks the ability to perceive this motion, making it ineffective under highly dynamic conditions. In this study, we enhanced the robotic US system by integrating motion awareness through a vision-haptic fusion motion control framework, as illustrated in Fig. 3. Specifically, the self-calibrating predictive model continuously updates the desired position of the probe. The position error between the desired and real-time positions is incorporated into the force-sensing admittance controller to dynamically adjust the movements of the probe

$$m_f \ddot{z_p}(t) + d_f \dot{z_p}(t) + k_f \left(z_p(t) - z_d(t)\right) = f_z(t) - f_d \tag{10}$$

where $m_f, d_f, k_f \in \mathbb{R}$ represent the mass, damping, and stiffness parameters, respectively, while $z_p(t)$ and $z_d(t)$ denote the real-time and desired positions of the probe tip in the z-axis of the robot base frame, respectively. The desired contact force f_d is set as a constant before scanning, and $f_z(t)$ represents the real-time contact force. To ensure accurate contact force measurement, a gravity compensation method proposed in [26] was employed to eliminate the effect of the probe's weight on the F/T sensor. The contact force is controlled by adjusting $z_p(t)$.

To enhance the robot's responsiveness and stability, the concept of MPC was employed. The self-calibrating predictive model estimates the position of the imaged object at the next control cycle, which serves as the desired position for the improved admittance controller to follow. During model recalibration, the real-time position $z_c(t)$, obtained from the vision-based motion monitoring system, is set as the desired position. Therefore, the desired position for the controller is defined as follows:

$$z_d(t) = \begin{cases} z_m(t+\Delta t), & \text{if } |z_m(t) - z_c(t)| < \lambda_z \\ z_c(t), & \text{if } |z_m(t) - z_c(t)| > \lambda_z \end{cases} \tag{11}$$

where Δt represents the control time interval. The velocity $\dot{z_p}(t)$ is obtained from (10). Subsequently, the error $e_k \in \mathbb{R}^6$ between the current and target probe pose is given by the following equation:

$$e_k = \left[d_x, d_y, \dot{z_p}(t), d_{rx}, d_{ry}, d_{rz}\right]^T. \tag{12}$$

The terms d_x and d_y denote the translational steps for scanning velocity and direction, whereas d_{rx}, d_{ry}, and d_{rz} represent the angular orientation changes. These parameters are established through trajectory planning and are independent of our control variables. Therefore, we utilized a preset scanning trajectory and parameters for evaluation in this work. The robot arm joints $\Delta\theta(t) \in \mathbb{R}^6$ are the control signal for the robot servo system, which is expressed as follows:

$$\Delta\theta(t) = \left(J^T(\theta(t))\, J(\theta(t)) + \lambda_\theta^2 I\right)^{-1} J^T(\theta(t))\, e_k \tag{13}$$

where $J(\theta(t)) \in \mathbb{R}^{6\times 6}$ represents the robot Jacobian matrix and λ_θ is a nonzero damping constant [27]. During the US acquisition, the proposed vision-haptic fusion motion control framework employs the visual module to predict the respiratory motion, enabling preemptive adjustments to probe movements. Concurrently, the haptic module monitors the contact force, ensuring stable interaction during motion changes.

E. 3-D US Imaging Compensation

To address respiratory motion-induced distortions in volume measurements, we propose a 3-D US imaging compensation framework that integrates position compensation and anatomical priors, as depicted in Fig. 4. Initially, the movement of the imaged object is monitored, facilitating the reconstruction in a relatively static frame. We then incorporate vessel anatomy to construct a probability distribution model for the vessel centerline, applying filtering techniques to further refine the results.

1) Position Compensation: To compensate for distortions caused by respiratory motion, the primary idea is to perform reconstruction within a frame where the imaged object remains relatively static. To achieve this, a fixed frame $\{O\}$ is established on the imaged object, and the vision-based monitoring system is used to continuously track its movements. During scanning, the calibration board is rigidly connected to the imaged object, enabling real-time calculation of the transformation matrix ${}^C_O\mathbf{T}$ from frame $\{O\}$ to the camera frame $\{C\}$.

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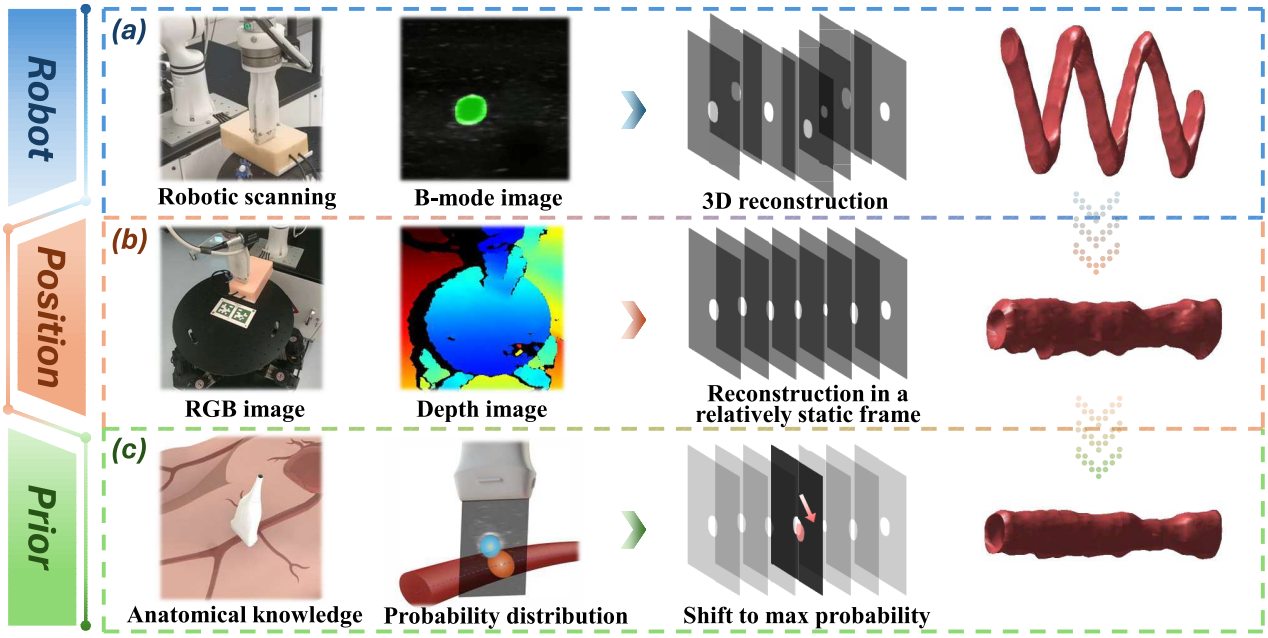


Fig. 4. Pipeline of the 3-D US imaging compensation framework consists of the following. (a) Acquisition of US images and volumes under respiratory motion. (b) Reconstruction in a relatively static frame using the vision-based motion monitoring system. (c) Establishing a probability distribution model of the vessel centerline using anatomical priors and calibrating the vessel mask by maximizing the product of the probability density functions of the observations and prior distributions.

Consequently, the pixel coordinates ${}^{I}\mathbf{P}$ are transformed into voxel coordinates ${}^{O}\mathbf{P}$ using the following equation:

$$ {}^{O}\mathbf{P} = {}^{O}_{C}\mathbf{T}\,{}^{C}_{B}\mathbf{T}\,{}^{B}\mathbf{P} = {}^{O}_{C}\mathbf{T}\,{}^{C}_{B}\mathbf{T}\,{}^{B}_{E}\mathbf{T}\,{}^{E}_{S}\mathbf{T}\,{}^{S}_{I}\mathbf{T}\,{}^{I}\mathbf{P} \tag{14} $$

where ${}^{C}_{B}\mathbf{T}$ is derived from hand–eye calibration, ${}^{B}_{E}\mathbf{T}$ is obtained from robot forward kinematics, and ${}^{E}_{S}\mathbf{T}$ and ${}^{S}_{I}\mathbf{T}$ are constants determined by the mechanical design. As discussed earlier, a temporal mismatch exists between ${}^{B}_{E}\mathbf{T}$ and ${}^{I}\mathbf{P}$. Incorporating mismatch time t_{mis} from temporal calibration, (14) can be revised as follows:

$$ {}^{O}\mathbf{P}\,(t) = {}^{O}_{C}\mathbf{T}\,(t)\,{}^{C}_{B}\mathbf{T}\,{}^{B}_{E}\mathbf{T}\,(t)\,{}^{E}_{S}\mathbf{T}\,{}^{S}_{I}\mathbf{T}\,{}^{I}\mathbf{P}\,(t + t_{\text{mis}})\,. \tag{15} $$

Using (15), a 3-D volume of imaged vessels can be acquired within a relatively static frame.

2) Anatomical Priors: In freehand US scanning, experienced clinicians can mentally reconstruct the 3-D volume of imaged arteries. Their anatomical knowledge provides crucial prior information about the direction and shape of imaged vessels, significantly aiding in accurate 3-D imaging. Due to location errors in the motion monitoring system, the volume after position compensation still deviates from the ground truth. To further refine this volume, this study replicates the expertise of clinicians by integrating anatomical priors into the 3-D imaging compensation framework.

Our previous research [28] developed a deep learning-based network to reconstruct a patient's skinned multiperson linear model (SMPL) with internal bone and artery models from RGB-D images. By integrating this network with hand–eye calibration, we can determine the anatomical priors, such as the direction and shape of the imaged artery, within frame $\{S\}$. In this study, cross-sectional imaging is performed along the x-axis of frame $\{S\}$. Consequently, the 2-D image is aligned with the anatomical volume using the x-coordinate values. The anatomical volume is then used to establish a probability distribution model of the vessel centerline. Due to individual anatomical variations, the anatomical information from the learning-based network is not absolutely accurate. Therefore, we assume that the distribution of the accurate vessel centerline points follows a 2-D Gaussian distribution $\mathcal{N}(\mu, \Sigma)$ within the yz plane. The mean μ of this distribution represents a point on the centerline, while the variance Σ reflects the shape of the vessel. Based on this assumption, the distribution from the current vessel mask is denoted as $\mathcal{N}_i(\mu_i, \Sigma_i)$, while the distribution from the anatomical volume is represented as $\mathcal{N}_p(\mu_p, \Sigma_p)$. By incorporating the principles of Kalman filtering, we integrate both the observation (the current US image) and the prediction (the anatomical priors) to estimate the most accurate position of the vessel centerline. The position μ_c is determined by maximizing the product of the probability density functions of the two Gaussian distributions

$$ \mu_c = \frac{\Sigma_i^{-1}\mu_i + \Sigma_p^{-1}\mu_p}{\Sigma_i^{-1} + \Sigma_p^{-1}}. \tag{16} $$

Because anatomical priors are only statistically reliable, adjustments are applied exclusively to the vessel centerline, leaving the vessel mask segmented from the US image initially unchanged. This mask is subsequently shifted by a distance of $\mu_c - \mu_i$.

IV. Experiments

A. Evaluation of Calibrations

To evaluate the accuracy of spatial calibration, a custom 3-D-printed, US-visible marker (Material Resin, Wenext,

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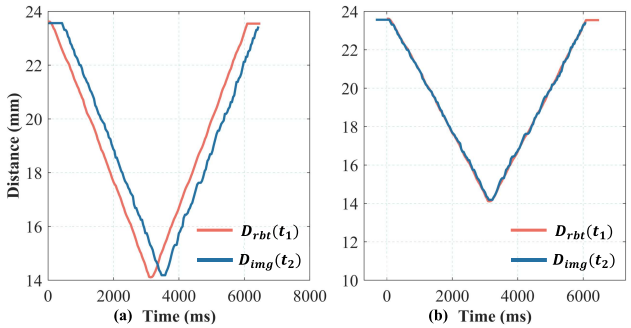


Fig. 5. Temporal calibration results. The red line represents distance from robot forward kinematics and the blue line represents distance from US images and spatial calibration. (a) Results before temporal mismatch compensation. (b) Results after temporal mismatch compensation.

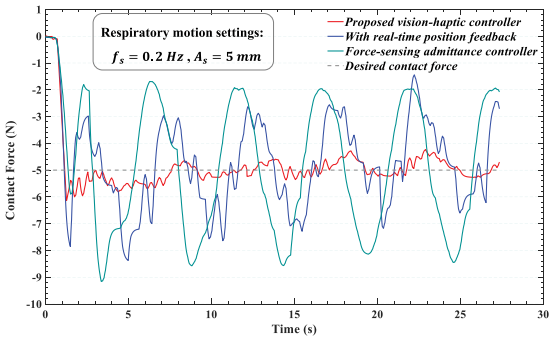


Fig. 6. Performance of the contact force tracking under respiratory motion (f_s = 0.2 Hz and A_s = 5 mm). The proposed vision-haptic fusion motion controller with predictive position feedback (red line). Admittance controller with real-time position feedback (blue line). Admittance controller [11] with only force sensing (green line).

China) was utilized. The marker's top was designed with a protuberance that could be easily identified in B-mode images to facilitate measurement. During the experiment, the marker was placed in a water-filled tank, and the coordinates of the protuberance in the robot base frame, denoted as M_{rbt}, were determined through mechanical design. Subsequently, the probe position was adjusted to ensure that the protuberance remained visible in the US images. Using the spatial calibration (1), the coordinates of the protuberance were transformed from the US image frame to the robot base frame, denoted as M_{img}. Data were collected at different probe positions and the Euclidean distance between M_{rbt} and M_{img} was calculated, resulting in a spatial calibration accuracy (mean $\pm$ SD) of 0.355 $\pm$ 0.018 mm. Additionally, the mean errors along the three axes were measured as follows: x-axis 0.154 mm, y-axis 0.235 mm, and z-axis 0.217 mm.

Furthermore, we evaluated the results of temporal calibration. To simplify the calibration process, the probe was controlled to move up and down along the z-axis of the robot base frame. During this process, the distance between the probe tip and the protuberance was calculated in real time using the robot forward kinematics, denoted as $D_{\text{rbt}}(t_1)$. Similarly, the same distance was determined using US images and the spatial calibration, denoted as $D_{\text{img}}(t_2)$. The time-dependent curves of the distances obtained from the two methods are shown in Fig. 5(a). It is evident that the distance calculated from the US images exhibits a temporal lag, with an average time of 352.9 ms. After compensating for this temporal mismatch, the experiment was repeated, as shown in Fig. 5(b). The results demonstrate a significant improvement in the temporal synchronization between the robot pose and the US images.

To evaluate the accuracy of the hand–eye calibration, a calibration marker was fixed in the robot base frame, and the coordinates of fixed points on the marker were determined through mechanical configuration. Meanwhile, the coordinates of these fixed points in the camera frame were obtained using the Intel RealSense Viewer software. The experimental results indicated that the mean coordinate error following the hand–eye calibration was 1.55 mm.

B. Evaluation of Robot Motion Control

To fully evaluate the proposed vision-haptic fusion motion control framework, a specialized experimental platform was devised, as shown in Fig. 2(a). Since the quality of US images is highly dependent on the contact force, we employed real-time contact force measurements to quantitatively evaluate the stability of US acquisition. A US vessel phantom with flat surface was mounted on a six-axis platform (LJ-6D-06, Nanjing Lingjin Automation Equipment, China), which was used to simulate respiratory motion. To facilitate evaluation, we utilized a preset scanning trajectory and parameters for the phantom to achieve full coverage imaging: a scanning velocity of 10 mm/s, a scanning direction along the vessel, and a vertical imaging angle. Considering human respiratory motion, the platform was driven to move up and down along the z-axis in a sinusoidal form and the desired contact force was set to 5 N. To effectively demonstrate the advantages of the improved admittance controller with visual position feedback, we compared it with a state-of-the-art admittance controller that relies solely on force sensing [11]. Additionally, to highlight the contribution of the predictive model, we tested a controller that uses only real-time positions obtained from the vision-based motion monitoring system as feedback. This method was the same as the control framework used during the predictive model recalibration. The respiratory amplitude was set to 5 mm with a frequency of 0.2 Hz, which aligns with the respiratory motion settings in [25].

The contact force tracking errors are shown in Fig. 6. After contact occurred, the proposed method maintained contact forces ranging from 4.26 to 6.14 N, while the real-time position feedback method exhibited contact forces ranging from 1.44 to 8.37 N. The force-sensing admittance controller performed the worst, with contact forces ranging from 1.68 to 9.15 N. The contact force tracking errors (mean $\pm$ SD) of these methods were 0.29 $\pm$ 0.21 N, 1.28 $\pm$ 0.83 N, and 2.10 $\pm$ 1.00 N. These results demonstrated that the proposed vision-haptic fusion motion control framework significantly enhances the system to maintain contact force and acquire stable US images under respiratory motion. However, we observed that using real-time position feedback did not achieve satisfactory results. This shortcoming is primarily due

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to the lack of temporal predictive capability, preventing the system from responding to movements in advance. Nevertheless, compared to the admittance controller relying solely on force sensing, the integration of visual position feedback still improved US acquisition stability under respiratory motion.

To further evaluate the performance of the proposed motion control framework, experiments were conducted under conditions of changing respiratory frequency. The respiratory amplitude was set to 5 mm, while the frequency was reduced from 0.4 to 0.2 Hz and then stopped. The deviations in contact force during the experiment are shown in Fig. 7. Based on the controller's response, the process was divided into five stages.

1) *Stage I:* The respiratory frequency was 0.4 Hz. During this stage, the predictive model accurately predicted the desired position, resulting in contact forces ranging from 4.22 to 6.17 N and a tracking error (mean ± SD) of 0.37 ± 0.23 N.
2) *Stage II:* The frequency suddenly decreased to 0.2 Hz. At this point, discrepancies arose between the predicted and actual movements. Real-time position feedback was used, and the self-calibrating predictive model recalibrated parameters. As a result, the contact force ranged from 2.15 to 6.94 N, with a tracking error of 1.12 ± 0.70 N.
3) *Stage III:* The frequency remained at 0.2 Hz. The parameters of the predictive model were successfully updated, enabling accurate prediction of the desired position. The contact force ranged from 4.16 to 5.47 N, and the tracking error was 0.34 ± 0.21 N.
4) *Stage IV:* Respiratory motion suddenly stopped, with the frequency dropping to 0. At this stage, discrepancies reappeared, and real-time position feedback was used again. The contact force ranged from 4.30 to 6.98 N, with a tracking error of 0.78 ± 0.72 N.
5) *Stage V:* Respiratory motion remained stopped, and the predictive model parameters were fully updated. The contact force ranged from 4.99 to 5.19 N, with a tracking error of 0.12 ± 0.05 N.

When the frequency decreases from 0.4 to 0.2 Hz, the predictive model takes longer to accurately recalibrate parameters compared to the frequency decrease from 0.2 Hz to stop. This is because the update requires data collection for fitting, and stationary conditions facilitate this process. Additionally, a noticeable jump in contact force occurred during frequency switching. When the motion status changed, the six-axis platform returned to its zero position before resuming movement at the new frequency. This rapid and unpredictable return-to-zero action was one of the primary reasons for the contact force tracking errors in Stages II and IV. Overall, the contact force ranged from 2.15 to 6.98 N, with a tracking error of 0.51 ± 0.52 N, demonstrating that the proposed robotic system effectively achieves stable US acquisition under dynamic respiratory motion.

To evaluate system latency, the update times for the key signals of the proposed motion control framework were measured. The mean (±SD) update times were 64.38 ± 7.84 ms for the visual signal, 4.61 ± 0.19 ms for the force signal, and 10.06 ± 0.06 ms for the robot control signal. Given the

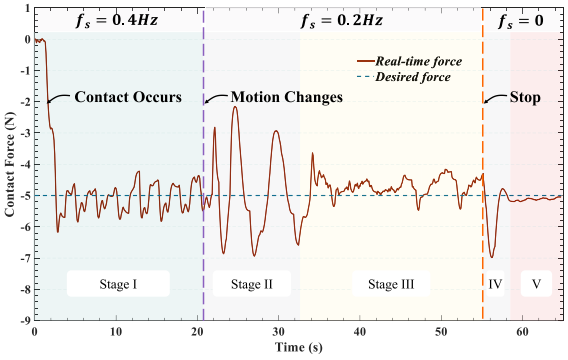


Fig. 7. Contact force tracking performance of the proposed motion control framework under respiratory frequency changes.

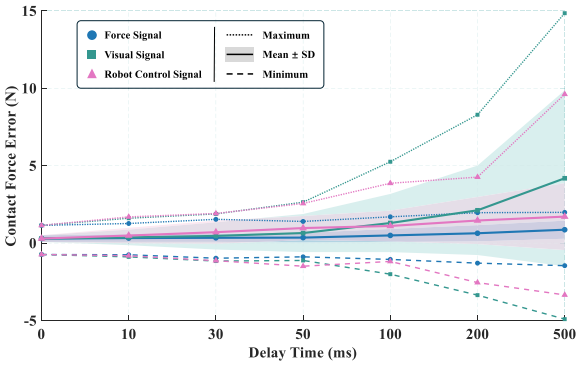


Fig. 8. Impact of visual, force, and robot control signal latency on contact force tracking performance of the proposed motion control framework.

100-Hz control frequency of the system, these results indicate an inherent latency in the visual signal, whereas force feedback is received in real time within each cycle. The impact of each signal latency on tracking accuracy was further investigated through respiratory tracking experiments with induced delays. The experimental results, as shown in Fig. 8, indicate that the contact force error increases with signal latency. Notably, for delays below 50 ms, the robot control signal resulted in the largest mean tracking error. However, visual signal latency caused the most significant performance degradation for delays exceeding 50 ms. This shift occurs because the predictive model can compensate for short delays but fails as latency exceeds 50 ms. Conversely, force signal latency had a minimal impact on overall performance. The visual loop compensates for the majority of respiratory motion, which helps maintain a small contact force error. Consequently, the force feedback loop acts as a minor corrective term, making it less sensitive to delay. Overall, the system demonstrates robust performance for signal latencies below 50 ms, maintaining the mean contact force error within 1 N.

The desired contact force of 5 N in our experiments is a representative value commonly used in robotic US research and is frequently referenced in studies involving vessel phantoms [29], abdominal phantoms [8], and clinical spine imaging [30]. However, the optimal contact force in clinical practice is not fixed. Sonographers adjust the force based on the specific anatomical region and diagnostic goals of the US

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TABLE I
PERFORMANCE OF THE PROPOSED AND OTHER METHODS UNDER VARIOUS RESPIRATORY MOTION SETTINGS

Respiratory motion settings	**0.2Hz 5mm**	**0.2Hz 7.5mm**	**0.2Hz 10mm**	**0.2Hz 12.5mm**	**0.3Hz 5mm**	**0.3Hz 7.5mm**	**0.4Hz 5mm**
Proposed motion control framework	(4.26, 6.14) * 0.29 ± 0.21 *	(3.13, 6.41) 0.57 ± 0.42	(3.68, 7.75) 0.59 ± 0.48	(3.41, 7.52) 0.85 ± 0.89	(4.24, 6.55) 0.40 ± 0.29	(3.88, 6.87) 0.58 ± 0.55	(3.17, 6.62) 0.49 ± 0.41
With real-time position feedback	(1.44, 8.37) 1.28 ± 0.83	(1.07, 9.41) 1.89 ± 1.07	(0.08, 11.89) 2.87 ± 1.57	(0.19, 14.85) 3.54 ± 3.57	(0.53, 10.54) 1.95 ± 1.24	(0.61, 13.32) 3.04 ± 3.23	(0.43, 11.12) 2.80 ± 1.52
Force-sensing admittance controller	(1.69, 9.16) 2.10 ± 1.00	(0.02, 17.44) 4.30 ± 2.24	(0.03, 19.34) 5.93 ± 5.46	× ×	(0.02, 13.75) 4.58 ± 2.04	(0.05, 18.13) 5.63 ± 5.25	(0.04, 18.48) 4.99 ± 4.57

∗ Accuracy: (Minimum, Maximum) values of the contact force, and the Mean ± SD of the contact force error.
×: The contact force exceeds the safety limit of 20 N, indicating that this method cannot perform scanning under the given setting.

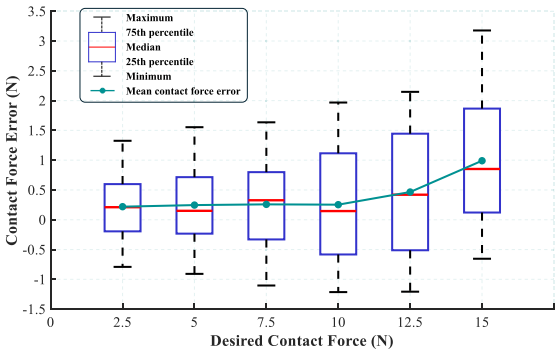


Fig. 9. Force tracking performance at various desired contact force.

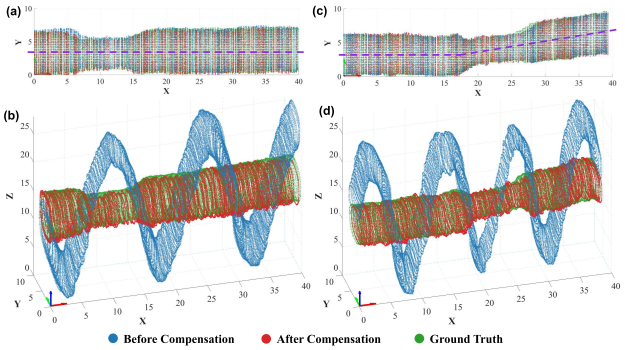


Fig. 10. (a) and (b) Performance of the 3-D US imaging compensation framework on a straight-line vessel. (c) and (d) Polyline vessel mimicking a human vascular bifurcation.

examination. The existing literature on robotic US systems indicates the commonly used contact force ranges, such as 2–4 N [31], 8 N [9], and 15 N [32]. To verify the ability of the proposed motion control framework to cover common safe force ranges, experiments were conducted with desired contact forces ranging from 2.5 to 15 N. As depicted in Fig. 9, the mean contact force error was maintained within 1 N, with the maximum error remaining below 3 N. A notable observation is the positive correlation between the contact force error and the desired contact force. This occurs because, at higher contact forces, the same probe displacement results in a larger change in contact force. Therefore, the error within the predicted position feedback loop increases, leading to an overall increase in force tracking error. In conclusion, the experimental results demonstrate that the proposed motion control framework can accommodate common scanning forces without causing patient discomfort or deforming the imaged target.

Additional experiments with various respiratory motion settings were conducted to further evaluate the performance of the motion control framework. The respiratory amplitude was set from 5 to 12.5 mm, covering the range of normal and deep respiratory motion for most individuals. The frequency was varied from 0.2 to 0.4 Hz, aligning with the typical human respiratory frequency range of 0.16–0.33 Hz. Each respiratory motion setting was repeated three times, with the statistical results presented in Table I. The proposed motion control framework consistently maintained a contact force tracking range within ±2 N, with an average error below 0.6 N, even when the frequency increased to 0.4 Hz or the amplitude reached 10 mm. At the maximum amplitude setting of 12.5 mm, the proposed framework achieved a peak error of 2.5 N and an average error of 0.85 N. In contrast, the admittance controller exceeded the 20-N safety threshold in all three repeated experiments under the same conditions, failing to perform the scan. These results demonstrate that the proposed motion control framework provides robust performance in tracking the desired contact force and acquiring stable US images under respiratory motion.

C. Evaluation of 3-D US Imaging Compensation

To qualitatively and quantitatively validate the 3-D US imaging compensation framework, the US volume of the vessel phantom was obtained using temporally aligned B-mode images and corresponding pose data. To effectively demonstrate the performance, the experiment was conducted with simulated respiratory motion (10-mm amplitude and 0.2-Hz frequency). Meanwhile, the movement of the phantom was tracked by the motion monitoring system. To intuitively present the results, the internal vessels were automatically segmented from cross-sectional US images using a pretrained U-Net model. The robotic pose data and tracking information were aligned through timestamps, allowing the corresponding vessel masks to be transformed into a relatively static frame $\{O\}$ for 3-D reconstruction through (14). The same method was applied to generate the 3-D volume of the imaged vessel

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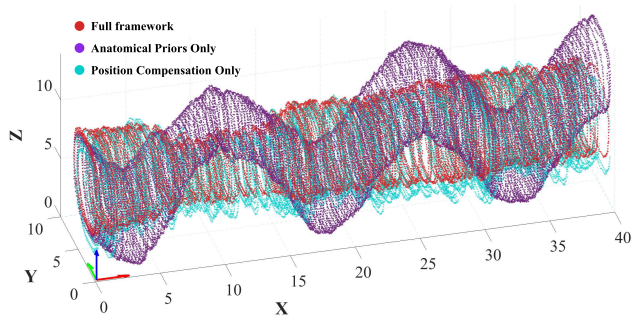


Fig. 11. Ablation study of the 3-D US imaging compensation framework.

under stationary conditions, which served as the ground truth. To validate the robustness of the compensation framework, the experiment utilized two vessel shapes: a straight line and a polyline simulating human vascular bifurcation. The distribution of these vessels, known from the phantom fabrication, served as anatomical priors. Therefore, a probability distribution model of the vessel centerline was established in the frame $\{O\}$, and the reconstructed volume was filtered using (16). To better compare and illustrate the performance, the vessel volume was also reconstructed without the 3-D compensation framework. The reconstructed volumes are shown in Fig. 10.

The results demonstrated that the compensation framework effectively addressed the distortions caused by respiratory motion. Additionally, the Chamfer distance (d_{CD}) was used for quantitative accuracy evaluation. The Chamfer distance measures the distance between two point clouds, S_1 and S_2, where smaller values indicate greater similarity between the point clouds. The calculation formula is as follows:

$$d_{\mathrm{CD}}(S_1, S_2) = \frac{1}{S_1}\sum_{x\in S_1}\min_{y\in S_2}\|x-y\|_2^2 + \frac{1}{S_2}\sum_{y\in S_2}\min_{x\in S_1}\|x-y\|_2^2. \tag{17}$$

The Chamfer distances between the uncompensated point cloud and the ground truth were 5.39 mm for the straight-line vessel and 5.73 mm for the polyline vessel. After compensation, these distances were significantly reduced to 0.48 and 0.53 mm.

To demonstrate the individual contributions of position compensation and anatomical priors within the framework, we compared the results of applying each component separately. As illustrated in Fig. 11, the volume using only position compensation exhibits noticeable surface irregularities, which are caused by system calibration and motion tracking errors. Conversely, employing anatomical priors alone proved insufficient to eliminate the primary respiratory motion-induced distortions. Quantitatively, the Chamfer distance between the point cloud reconstructed using only position compensation and the ground truth was 0.98 mm. Using only anatomical prior compensation, this distance increased to 2.26 mm. Both error metrics are larger than the 0.48 mm achieved by the proposed full framework. These results demonstrate that the two components serve complementary functions. Position compensation primarily corrects distortions induced by respiratory motion, while anatomical priors refine the surface geometry. Thus, the integration of both components in the 3-D US imaging compensation framework is crucial for achieving accurate volume measurements.

Furthermore, we performed additional evaluations of the compensation framework under varying respiratory motion conditions. The amplitude was set to 5 mm, with frequencies of 0.3 and 0.4 Hz. At a frequency of 0.3 Hz, the Chamfer distance before compensation was 3.27 mm, which was significantly reduced to 0.67 mm after compensation. Similarly, at a frequency of 0.4 Hz, the CD decreased from 2.41 to 0.89 mm after compensation. These results demonstrate that the proposed compensation framework effectively reduces respiratory motion-induced distortions in volume measurements. This improvement has important clinical implications for the diagnosis of vessel diseases, as it facilitates the measurement of small volume changes.

V. Discussion and Conclusion

Respiratory motion presents significant challenges for stable US acquisition and accurate volume measurements, limiting its reliability in cardiovascular, abdominal, and spinal clinical diagnoses. This study addresses these limitations by developing a respiratory motion-robust robotic 3-D US system. First, the system employs a vision-based motion monitoring system to construct a self-calibrating predictive model that dynamically estimates respiratory motion and predicts the desired contact position. And then the vision-haptic fusion motion control framework integrates predicted position feedback into the force-sensing admittance controller. Furthermore, the controller is enhanced with model predictive capabilities by the self-calibrating predictive model. This approach achieves tracking errors of 0.29 ± 0.21 N and maintains contact forces between 4.26 and 6.14 N under respiratory motion (5-mm amplitude and 0.2-Hz frequency). Consequently, the system is able to acquire stable, high-quality US images under respiratory motion for volume reconstruction. Second, the 3-D compensation framework reduces respiratory motion-induced distortions in volume measurements by reconstructing volumes in a relatively static frame and refining results using anatomical priors. The experimental results demonstrate that the compensation framework reduces the Chamfer distance between the reconstructed and the ground truth volumes from 5.4 to 0.48 mm.

However, several limitations of the proposed system require further discussion. First, this work employs a sinusoidal model with dynamically updated parameters to describe respiratory motion and address frequency variations. However, this approach is restricted to the vertical component of the motion. In clinical practice, respiratory motion is often irregular, requiring more sophisticated modeling techniques. Second, the rigidly connected calibration board used for 6-DOF localization of the imaged object may limit the clinical applicability, as human bodies exhibit flexibility. These physical markers could potentially be replaced by advanced, marker-less tracking methods. For instance, techniques like FoundationPose [33] leverage CAD models or reference images for 6-DOF tracking. Registration-based methods also present a promising approach for motion tracking [20]. These advanced approaches

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could provide the detailed motion information required by our framework. Another limitation is the constrained application of anatomical priors. While effective for regular, statistically consistent structures like blood vessels, their utility diminishes when applied to complex and soft organs, such as the liver. Significant interpatient variability makes constructing an accurate probability distribution model challenging. In addition, US imaging requires the application of contact force to acquire images. Although our motion control framework reduces contact force variations during scanning, tissue deformation caused by preset probe pressure remains an unaddressed challenge that requires further compensation in future work.

In conclusion, this study presents a respiratory motion-robust robotic 3-D US imaging system. The primary contribution is the synergistic integration of three key elements: a self-calibrating predictive model for respiratory motion, the incorporation of predicted position feedback into the force-sensing admittance controller, and the first use of anatomical priors for US volume compensation. Such improvements we believe represent a critical step toward enabling the application of robotic US acquisition and volume measurements in clinical diagnoses affected by respiratory motion.

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